Step 12 Four-ROI Continuous Abstractness Analysis

This side analysis reused the Step 10/11 cohort and trial table, recomputed the variable-window HbO outcome for four user-defined ROIs, and fit the Step 11 continuous abstractness model separately for each ROI. Holm correction was applied across the four primary ROI tests.

ROI Definitions

ROI	Interpretive label	HbO pairs
Anterior Dorsal	Control-network / superior frontal cortex candidate	S1_D6, S1_D3, S3_D3, S5_D3, S5_D6
Anterior Ventral	Inferior frontal / semantic processing candidate	S2_D4, S2_D6, S5_D4, S4_D4, S5_D7
Posterior Dorsal	Superior parietal / visuospatial candidate	S3_D7, S3_D1, S8_D1, S8_D5, S6_D1
Posterior Ventral	Wernicke / semantic-multisensory candidate	S7_D2, S7_D5, S6_D2, S4_D2, S4_D7, S6_D7

ROI
Anterior Dorsal
Anterior Ventral
Posterior Dorsal
Posterior Ventral

ROI Model Results

ROI	Trials	Mean channels	Beta	SE	Raw p	Holm p	95
							Anterior
1075	4.92	1.642e-08	1.042e-08	0.125957	0.503828	[-4.894e-09, 3.774e-08]	
Posterior Dorsal	1075	4.55	1.152e-08	7.646e-09	0.142819	0.503828	[-4.121e-0 2.716e-08
Posterior Ventral	1075	5.94	7.335e-09	8.357e-09	0.387319	0.774638	[-9.757e-0 2.443e-08
Anterior Dorsal	1075	4.58	7.658e-09	1.412e-08	0.591687	0.774638	[-2.122e-0 3.654e-08

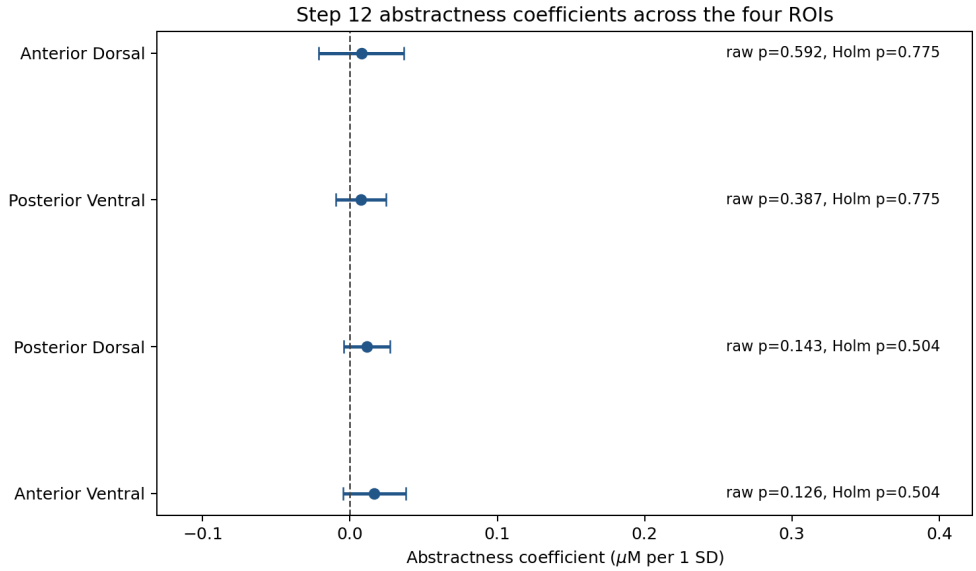

Figure 1: Continuous abstractness coefficients and 95% confidence intervals for the four ROI-specific models.

Descriptive Waveforms

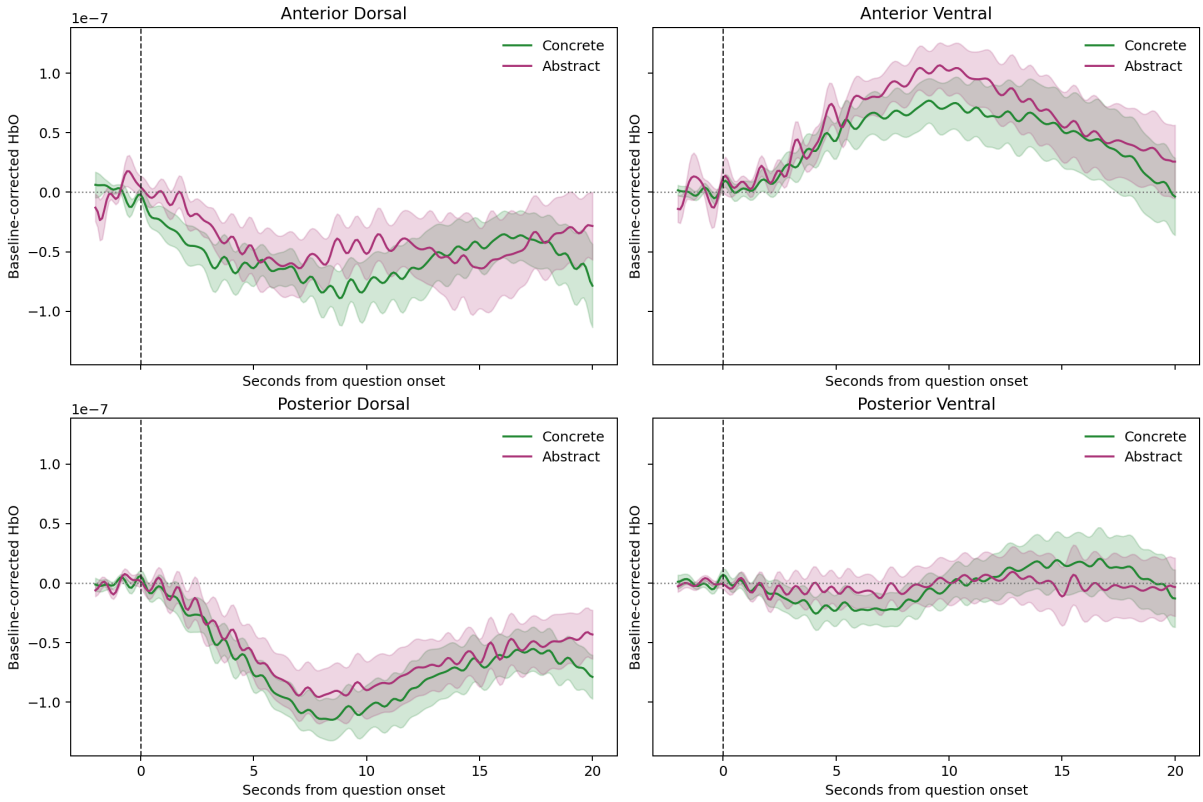

Figure 2: Onset-locked descriptive HbO waveforms for the four ROI definitions, averaged within participant-session and then across the Step 10/11 cohort.

Conclusion

Using the Step 10/11 trial table and the Step 11 continuous abstractness model separately within four new ROIs, none of the ROI-specific abstractness effects survived Holm correction across the four tests. The smallest raw p-value was in Anterior Ventral (beta=1.642e-08, SE=1.042e-08, raw p=0.125957, Holm p=0.503828).